Abstract

Introduction

As global energy demand continues to rise and the transition toward renewable energy becomes increasingly important, improving the efficiency of existing solar technologies is essential. Solar photovoltaic (PV) systems offer a promising means of generating clean electricity, yet the amount of energy they capture depends on several environmental and geometric factors, including solar irradiance, weather conditions, and the angle at which sunlight strikes the panel. Conventional fixed-orientation solar panels cannot continuously adapt to the Sun's changing position throughout the day, potentially limiting the amount of solar energy they collect.

Solar tracking systems address this limitation by adjusting a panel's orientation to follow the Sun's apparent movement across the sky. In particular, solar geo-tracking uses geographic coordinates, time, and solar-position calculations to estimate the Sun's azimuth and elevation angles, enabling panels to adjust their orientation accordingly. By dynamically aligning panels with the direction of sunlight, these systems have the potential to increase energy capture relative to fixed-panel configurations. However, the effectiveness of tracking depends on factors such as positional accuracy, mechanical response, environmental conditions, and the energy consumed by the tracking mechanism itself.

This paper investigates the potential of solar geo-tracking to improve solar energy capture through a system that calculates the Sun's position and adjusts panel orientation accordingly. By examining solar-position predictions, tracking behavior, and energy-capture performance, the project aims to evaluate the way dynamic panel alignment can reduce losses associated with suboptimal orientation.

known as the Shockley-Quiesser limit which states that the maximum theoretical efficiency of a standard single-junction solar cell stands at approximately 33.7%. The existence of the theoretical ceiling highlights an important distinction between improving a solar cell's intrinsic conversion efficiency and increasing the total electrical energy generated by a photovoltaic system. Material innovations, such as improved semiconductor structures, surface texturing, and anti-reflective coatings can reduce losses occurring within the cell itself. Increasing the amount of sunlight that reaches the panel at a favorable angle contains another opportunity of improving overall energy prodcution. Even a highly efficient photovoltaic cell cannot generate its maximum possible output when the incident solar radiation is substantially reduced or when the panel is poorly oriented relative to the available sunlight. This creates an opportunity to investigate solar tracking as a complementary approach to improving system-level performance.

Literature Review

Geometry of Sunlight

The power generated by a photovoltaic panel depends strongly on the intensity of the sunlight incident upon its surface. Solar irradiance, describes the instantaneous solar power received per unit area. The total energy generated over time depends on the irradiance available throughout the day, the panel's conversion efficiency, its area, and the performance of the electrical system.

The orientation of a solar panel determines how effectively it intercepts incoming direct sunlight. When sunlight strikes a panel perpendicular to its surface, the panel receives the greatest possible direct-beam irradiance for a given solar beam intensity. As the angle between the incoming sunlight and the panel's surface normal increases, the same beam of sunlight is distributed across a larger effective area, reducing the power received per unit panel area. In an ideal model, [ ]. This relationship assumes that direct sunlight is available and that the panel is not shaded; actual panel output also depends on diffuse radiation, reflection, temperature, and electrical characteristics.

The angle of incidence changes continuously because of Earth's rotation and its orbit around the Sun. Solar elevation, which describes the Sun's angular height above the horizon, and solar azimuth, which describes its direction along the horizon, vary with the time of day, date, latitude, and longitude. Seasonal changes in Earth's axial orientation also alter the Sun's apparent path across the sky. Consequently, an orientation that is favorable during one part of the day or year may become less favorable later. It is also important to note that the sun’s path in the sky varies across different regions of the Earth. This study presents data from this city of San Jose, California in the northern hemisphere of the Earth.

Fixed-orientation photovoltaic systems use a stationary tilt and azimuth selected to provide suitable annual or seasonal performance. This design is mechanically simple, relatively inexpensive, and often effective because photovoltaic panels can collect both direct and diffuse sunlight. Depending on its location and orientation, it may therefore receive less direct sunlight during portions of the day than a panel that actively changes its position. Solar tracking attempts to address this geometric limitation by dynamically adjusting the orientation of a panel to follow the apparent movement of the Sun.

Solar Geo-Tracking as an Approach to Improving Energy Capture:

Solar tracking systems use mechanical and computational methods to adjust the orientation of photovoltaic panels throughout the day. Single-axis trackers generally rotate a panel around one axis, while dual-axis trackers can adjust both azimuth and elevation to follow the Sun more closely. The potential advantage of a tracking system depends on geographic location, seasonal solar

trajectories, local weather, installation costs, and the characteristics of the photovoltaic installation. Tracking may be particularly useful at sites where direct sunlight is abundant, although its benefits can be smaller under predominantly diffuse-light conditions.

Solar geo-tracking introduces a computational approach to determining the Sun's position. Rather than relying exclusively on light sensors to detect the brightest direction, a geo-tracking system can use geographic coordinates and date-and-time information to calculate the Sun's expected azimuth and elevation. A solar-position algorithm then estimates the appropriate panel orientation. A controller can translate these calculated angles into mechanical movements using actuators such as servo motors.

The accuracy of the resulting alignment depends on several factors, including the accuracy of the time and geographic coordinates, the mathematical model used to calculate solar position, the calibration of the tracking mechanism, and the correspondence between calculated solar angles and the physical orientation of the panel. Mechanical backlash, actuator limitations, power consumption, and sensor or control errors can prevent the panel from reaching its intended orientation.

(add error analysis)

Importantly, accurately tracking the Sun does not automatically guarantee a proportional increase in electrical energy production. The relationship between orientation and output also depends on the available direct and diffuse irradiance, the panel's temperature, and the electrical response of the photovoltaic system. Tracking also introduces additional hardware and energy requirements. Motors, controllers, and sensors consume electricity, and moving components may require maintenance over time. A useful assessment must therefore consider not only the energy collected by a tracking panel, but also the additional energy and resources required to operate the tracking system.

Environmental Constraints

Although panel orientation is an important factor in solar energy capture, environmental conditions can substantially alter photovoltaic output. Solar resources vary across the world according to latitude, atmospheric composition, cloud cover, moisture, aerosols, elevation, and seasonal weather patterns. The World Bank's global assessment of photovoltaic potential identifies arid and semiarid regions in parts of Africa, the Middle East, Australia, and the Americas as areas with particularly strong solar resources, while frequent cloud cover, atmospheric aerosols, and lower seasonal solar angles can reduce the available resource in other locations [ ]. These differences demonstrate why a solar technology that performs effectively under one set of environmental conditions may produce different results elsewhere.

Cloud cover and atmospheric scattering influence the quantity and direction of solar radiation reaching a panel. Thick clouds can substantially reduce direct sunlight, while thin or broken clouds may produce rapidly fluctuating irradiance. Clouds also scatter sunlight, increasing the diffuse component of radiation. Because diffuse sunlight arrives from a range of directions rather than predominantly from the solar disk, orienting a panel directly toward the calculated solar position may provide a smaller advantage under heavily overcast conditions than under clear skies. Some cloud configurations can also temporarily increase irradiance through reflection and scattering, illustrating that cloud effects are not always uniform. The performance of a tracking system should therefore be evaluated under different cloud conditions rather than assuming that every improvement in alignment produces an equivalent improvement in energy generation.

Temperature is another important constraint. Although sunlight provides the energy required for photovoltaic conversion, high operating temperatures generally reduce the electrical efficiency of conventional silicon photovoltaic modules. As a cell heats up, its voltage typically decreases, reducing the electrical power it can deliver under otherwise comparable conditions. The magnitude of this reduction depends on the specific module's temperature coefficient and operating characteristics. A panel installed in a hot, sunny desert may receive intense solar radiation while simultaneously experiencing elevated temperatures that reduce its conversion efficiency. In contrast, a cooler environment with comparable irradiance may allow the same module to operate more efficiently. This illustrates why high ambient temperatures and high solar energy availability should not be treated as interchangeable measures of photovoltaic performance [].

Dust, sand, soot, and other airborne particles can further reduce output by accumulating on panel surfaces and obstructing incoming sunlight. Soiling is particularly relevant in arid environments, where rainfall may be infrequent and dust accumulation can persist for extended periods. The National Renewable Energy Laboratory has documented how accumulated particles reduce photovoltaic performance and how local rainfall patterns and the characteristics of deposited dust affect the effectiveness of natural cleaning [5]. Although some precipitation can remove surface contaminants, rainfall does not guarantee complete cleaning, and certain deposits can adhere strongly to the panel. A tracking system cannot directly eliminate these optical losses; its effectiveness must be considered alongside cleaning requirements and local maintenance conditions.

Humidity, precipitation, snow, and severe weather introduce additional variability. Atmospheric moisture and associated clouds can reduce direct solar radiation, while rain may temporarily decrease output through cloud cover even as it helps remove dust. Snow can obstruct the panel surface, preventing sunlight from reaching the cells, although snow-covered surroundings can also reflect additional light onto exposed panels. Wind can cool a panel and partly offset temperature-related losses, but strong winds and storms can place mechanical stress on mounting structures and tracking components. Hail, debris, and other severe weather can damage equipment and cause interruptions in electricity generation. The net effect of each condition depends on the local environment, panel design, installation, and duration of exposure.

Geographic variation is therefore central to evaluating solar-tracking technology. In sunny, dry regions with abundant direct radiation, tracking may offer meaningful opportunities to increase the amount of sunlight intercepted throughout the day, although dust and heat can offset some of the gains. In humid tropical regions, frequent cloud cover can reduce the advantage of following the direct solar beam. In higher-latitude regions, shorter winter days and lower solar elevation can change the value of tracking across seasons. In locations with highly variable cloud cover, the potential benefit may fluctuate considerably from one hour to another. These regional differences suggest that tracking performance should be understood as a function of both solar geometry and local environmental conditions, rather than as a fixed improvement that applies equally to every installation.

Objectives

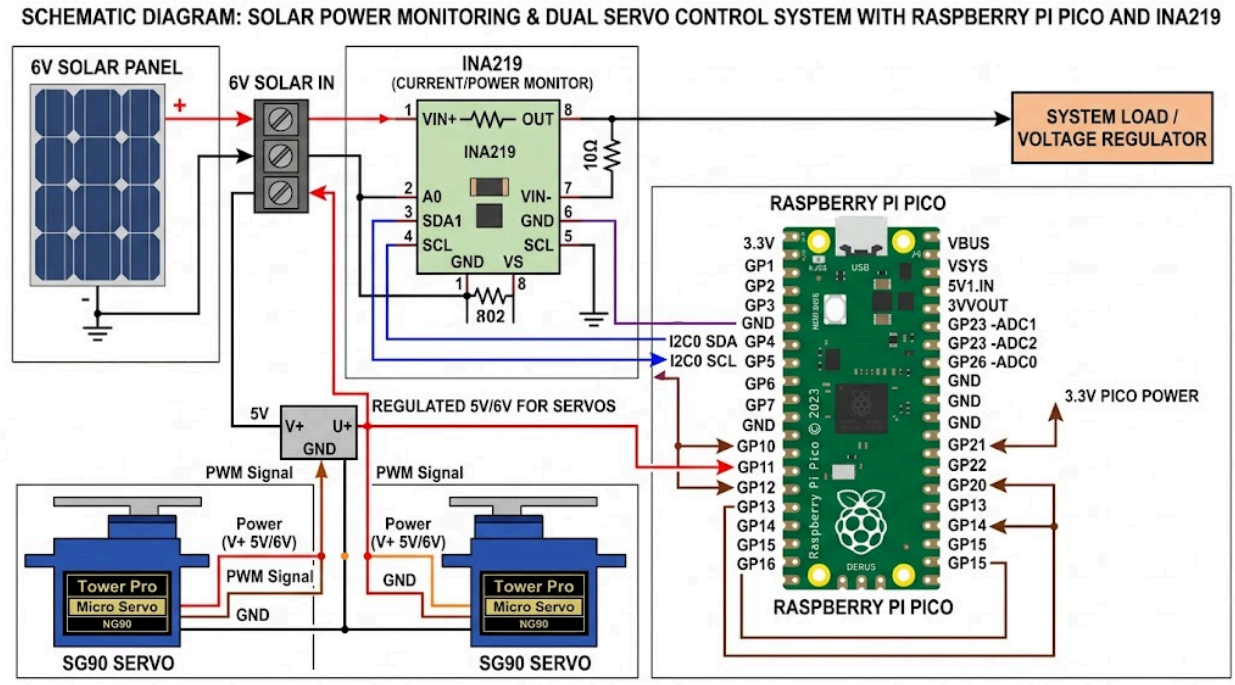
The constraints described above indicate that increasing photovoltaic energy production requires more than improving the conversion properties of solar cells alone. In the past, hotovoltaic panels integrated with geo-trackers have been well-documented in existing literature, yet standard systems often overlook the net-energy and net-cost balance of these systems. This study bridges that gap by analyzing tracking while evaluating the internal power consumption required to operate the tracking mechanism. This research investigates the design and implementation of a cost-effective solar geo-tracking system that utilizes geographic coordinates and solar-position calculations to adjust photovoltaic panel orientation in real time, with the objective of maximizing energy capture while minimizing system costs.

Hypothesis

I hypothesized that a dual-axis solar geo-tracking system adjusting a photovoltaic panel's orientation according to the Sun's calculated azimuth and elevation will produce greater electrical power output than when it is at a fixed reference orientation. Furthermore, my experiment measures the power at various orientations around the exact direction of the sun, to find out a direction that provides the most power efficienct. My hypothesis is that searching for the optimal direction will provide further efficiency gains.

Methodology

Figure 1: schematic of whole system



The system used for this research was developed using a Raspberry Pi Pico microcontroller programmed in MicroPython (Figure 1). The system rotates by a dual-axis servo mechanism built with two 180 degree servos (Figure 1). Our prototype uses a 6 volt, watt photovoltaic panel (Figure 1) with two servos controlling its horizontal and vertical orientation. An INA219 current and voltage sensor (Figure 1) communicates with the Raspberry Pi Pico through the Inter-Integrated Circuit communication protocol. The sensor provides the electrical measurements necessary to evaluate the panel's performance at different orientations.

Figure 5: temp and humidity sensor schematic

Additionally, the system incorporates a DHT22 temperature and humidity sensor (Figure 5) to record ambient environmental conditions. A separate ADC converter is implemented to estimate panel voltage using a resistor voltage divider. These components collectively enable the system to calculate solar position, adjust panel orientation, collect electrical measurements and record experimental data.

The solar position algorithm calculates the Sun's azimuth and elevation using geographic coordinates and hard-coded date and time information. Latitude and longitude specify the observation location, while the date and time establish the temporal reference for the calculation. The implementation uses astronomical equations to determine the Sun's apparent position

relative to the Earth First the program converts the date and time into a Juilian Day, a continuous numerical representation of time used in astronomical calculations. The Julian Day is subsequently used to calculate the Julian century relative to the J2000.0 reference epoch. These values provide the temporal inputs for determining the Sun's geometric mean longitude, mean anomaly, and orbital eccentricity. Corrections are then applied to account for the elliptical nature of Earth's orbit and the apparent position of the Sun, including adjustments to its ecliptic longitude and the obliquity of the ecliptic. Using these quantities, the program calculates solar declination and right ascension. Solar declination describes the angular position of the Sun north or south of the celestial equator, while right ascension specifies its position along the celestial coordinate system. The algorithm then calculates Greenwish Mean Sidereal Time and converts it into Local Sidereal Time using the observer's longitude. The resulting local sidereal time and solar right ascension determine the hour angle, which represents the Sun's angular displacement from the local meridian.

Finally the solar declination, hour angle, and geographic latitude are used to calculate solar elevation and azimuth. Elevation represents the Sun's angular height above the horizon, whereas azimuth represents its horizontal direction. These calculated coordinates provide the reference values for positioning the solar panel. The implementation includes time-conversion and sidereal-time calculations that must be verified against a validated astronomical reference before the resulting angles are treated as accurate experimental predictions.

The calculated solar coordinates are translated into physical movement through two servo motors controlled by the Raspberry Pi Pico. The servo at the base is connected to GPIO pin 15 and is intended to control horizontal movement, while the second servo is connected to GPIO pin 14 and controls vertical movement. Because the servos operate within mechanical limits, calculated solar coordinates cannot necessarily be translated directly into identical physical angles. The positioning mechanism therefore requires calibration to establish the correspondence between the command of angles sent to the servo and the panel's actual orientation. The accuracy of this mapping is critical because discrepancies between the calculated and physical angles may affect the validity of the comparisons between solar positions and measured electrical output.

The electrical performance of the photovoltaic panel is evaluated by measuring its voltage and current and calculating its electrical power output. The INA219 sensor connected through the computer interface measures current and calculates panel voltage through measurements of [*Fix*]]. Electrical power is then calculated using the relationship P = VI, where P represents electrical power in Watts, V represents panel voltage in volts, and I represents current in amperes.

To reduce the influence of short term measurement fluctuatjons, the program collects five consecutive measurements, with a 0.3 second interval between readings. The individual voltage, current, and power measurements are accumulated and averaged to produce a representative value for each measurement position.

A separate voltage-measurement function uses the Raspberry Pi Pico's ADC input on GPIO pin 26. The measured ADC value is converted into voltage using the microcontroller's 3.3 volt reference, after which the voltage divider's effect is applied to estimate the panel voltage. The two measurement methods provide separate approaches to voltage measurement; however, they should be cross-checked for consistency.

Algorithm:

Our experimental procedure investigates how changes in panel orientation affect electrical power output. The program first establishes a reference orientation at 250 degrees azimuth and 40 degrees elevation, the typical angle of rigid rooftop photovoltaic panels that collect sunlight parallel to the direction of the roof, specific to San Jose, California.

The panel is first moved to this reference position and the system pauses to allow the mechanical assembly to settle. An initial electrical measurement is then collected to provide a baseline for comparison with subsequent orientations.

The scanning procedure uses nominal angular increments of 15° in both azimuth and elevation. The program initializes the azimuth scan at a position 15° below the supplied solar azimuth and performs three iterations of horizontal positioning. At each horizontal position, the program evaluates three elevation positions, beginning 15° below the supplied solar elevation and increasing the elevation commanded to the servo by 15° after each measurement. Before each measurement, the program commands the servos to the designated orientation and introduces a one-second delay. It then obtains voltage, current, and power readings through the INA219 sensor. The resulting dataset records the commanded azimuth, commanded elevation, measured voltage, measured current, and calculated electrical power for each tested position.

The procedure therefore produces measurements from a reference orientation and a series of nearby positions arranged in a nominal three-by-three angular scan. The precise physical orientations represented by these commands depend on the servo calibration and angle transformations implemented in the control functions. The experimental records should be checked to confirm the actual orientations tested and to identify any missing or invalid measurements. By comparing the electrical output across the tested positions, the experiment can investigate whether nearby change local comparison rather than a comprehensive search across every possible panel orientation.

Algorithm and Code:

The code for the solar-tracking system was developed by researching existing open-source libraries and integrating them into a single MicroPython program for the Raspberry Pi Pico. Separate sections of the code were used to calculate the Sun's azimuth and elevation, control the servo motors, and collect voltage and current measurements from the INA219 sensor. These components were combined in a single file, main.py, which runs when the Pico is powered on.

Integrating the different functions into one program allowed the calculated solar position, motor movements, and electrical measurements to be used together during the experiment.

When the system starts, the panel is placed at a reference orientation, and its voltage and current are recorded using the INA219 sensor. The program then uses the location's latitude and longitude, along with the date and time, to calculate a Julian Day, which is used in determining the Sun's azimuth and elevation. In the current prototype, the geographic coordinates and time are entered manually, and the panel's reference alignment is established manually using a compass. These steps could be automated in a future version using a GPS module, a real-time clock, and a magnetometer, respectively.

After calculating the Sun's position, the program evaluates nine additional orientations surrounding the calculated solar direction. These positions are generated by varying both azimuth and elevation by 0° and ±15° relative to the calculated angles. The resulting combinations form a three-by-three grid of orientations, with the calculated solar direction at the center. At each position, the servos adjust the panel's orientation, after which the system records voltage and current and calculates electrical power using P = VI. Together with the initial reference measurement, these readings allow output at the fixed reference orientation to be compared with output at the calculated Sun direction and nearby orientations. This local scan also provides a way to investigate whether a small adjustment away from the calculated solar direction produces a higher measured output under the conditions of the experiment.

The program also records environmental conditions using a DHT22 temperature and humidity sensor. The geographic coordinates, date, time, temperature, and humidity are written to the output file alongside the ten sets of electrical measurements and their corresponding azimuth and elevation angles. Recording these values together makes it possible to compare power output across orientations while retaining information about the conditions under which each set of measurements was collected. The scan provides a local comparison rather than a comprehensive search across every possible panel orientation.

Environmental Conditions Monitoring

Environmental conditions are monitored to provide additional context for interpreting the electrical measurements. The DHT22 sensor, connected to GPIO pin 16, measures ambient temperature and relative humidity. The program requests these measurements before performing the orientation scan and records the results alongside the experimental metadata.

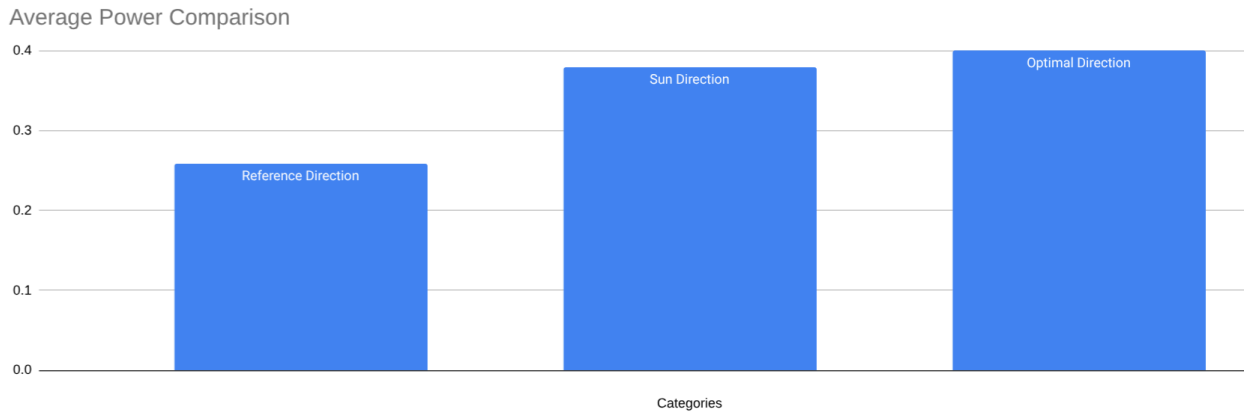
The recorded location and date and tiime provide a reference for the Sun's predicted position during each experiment. These variables are important because solar elevation and azimuth change throughout the day and across seasons. Ambient temperature, cloud cover, and variations in incident solar irradiance may also influence the electrical output of a photovoltaic panel.

The experimental setup uses the same nominal panel and measurement hardware across the positions within a scan. Nevertheless, the current implementation does not directly measure incident solar irradiance or actively control for changing weather conditions. Temperature and humidity measurements alone cannot fully characterize the environmental conditions affecting panel performance. Consequently, environmental variability should be acknowledged when interpreting differences between measurements, particularly when the scan takes long enough for sunlight conditions to change.

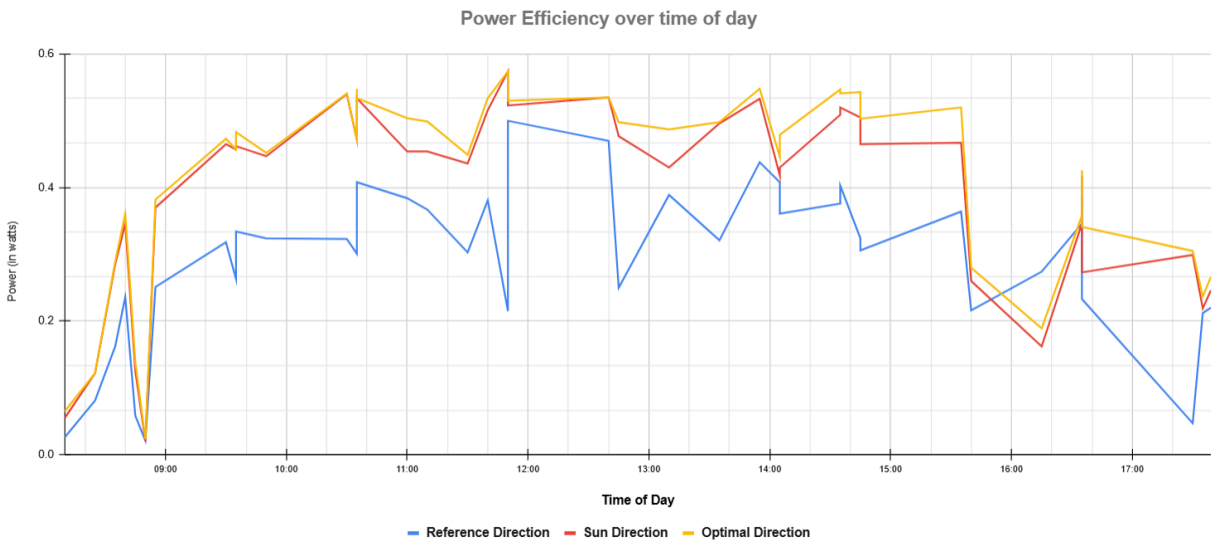
Data and Analysis

		Reference Direction		Sun Direction		Best Direction		Power (in watts)		
Date	Time	Azimuth	Elevation	Azimuth	Elevation	Azimuth	Elevation	Reference Direc	Sun Direction	Best Direction
10/6/2026	08:10	165	45	105.7	11.4	90.7	0	0.026	0.055	0.065
10/10/2026	08:25	165	45	109.48	13.32	109.48	13.32	0.081	0.122	0.122
10/11/2026	08:35	165	45	111.51	14.98	111.51	0	0.162	0.285	0.289
10/18/2026	08:40	165	45	114.31	14.65	129.31	2.65	0.236	0.346	0.361
10/5/2026	08:45	165	45	111.3	18.2	111.3	8.2	0.058	0.125	0.138
10/7/2026	08:50	165	45	112.41	30.76	127.41	33.76	0.021	0.02	0.023
10/16/2026	08:55	165	45	116.64	17.56	131.64	5.56	0.251	0.37	0.382
10/10/2026	09:30	165	45	121.32	24.98	121.32	9.98	0.318	0.465	0.473
10/11/2026	09:35	165	45	122.66	25.59	122.66	25.59	0.263	0.457	0.457
10/18/2026	09:35	165	45	124.58	24.17	124.58	12.17	0.334	0.462	0.483
10/12/2026	09:50	165	45	126.11	27.81	126.11	39.81	0.324	0.447	0.452
10/10/2026	10:30	165	45	134.65	34.39	134.65	19.39	0.323	0.54	0.541
10/9/2026	10:35	165	45	120.57	20.36	135.57	35.36	0.301	0.472	0.472
10/11/2026	10:35	165	45	136.22	34.81	121.22	22.81	0.35	0.539	0.548
10/18/2026	10:35	165	45	138.03	33.13	138.03	33.13	0.408	0.534	0.534
10/12/2026	11:00	165	45	143.11	37.73	128.11	37.73	0.384	0.454	0.504
10/7/2026	11:10	165	45	144.08	40.72	129.08	25.72	0.367	0.454	0.499
10/10/2026	11:30	165	45	151.25	41.59	151.25	26.59	0.303	0.436	0.449
10/18/2026	11:40	165	45	156.03	40.18	141.03	28.18	0.381	0.516	0.534
10/11/2026	11:50	165	45	157.8	42.96	157.8	42.96	0.215	0.574	0.574

For each experimental session, the program records the geographic coordinates, date, time, temperature, and humidity of 9 different orientations around the reference position. The resulting dataset allows electrical measurements to be compared across the tested orientations. We collected data readings across 3 days, every hour between 8:30 AM and 4:30 PM and observed the change across the orientations around the reference position at each hour.

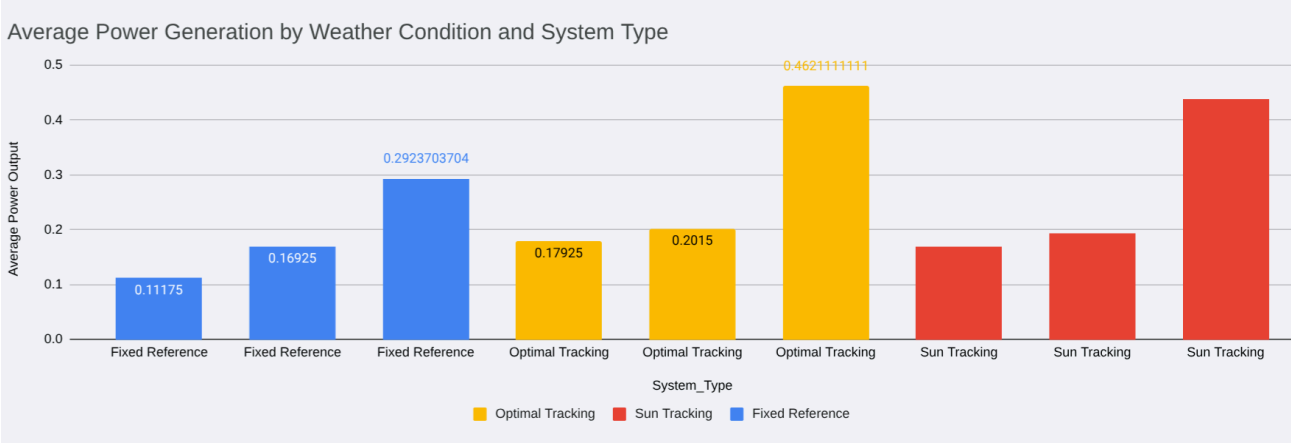


To examine the potential benefit of adjusting the panel's orientation, the average power values obtained for the reference direction, calculated Sun direction, and locally optimal direction were used to estimate energy production over an extended period. The average power values were 0.290 W for the reference direction, 0.389 W for the Sun direction, and 0.413 W for the locally optimal direction. Using these averages, the estimated energy output can be calculated by multiplying average power by the duration of operation. Over an assumed nine-hour period, the corresponding estimates are 2.61 Wh, 3.50 Wh, and 3.72 Wh, respectively. Relative to the reference direction, the estimated additional energy is therefore 0.89 Wh for the calculated Sun direction and 1.11 Wh for the locally optimal direction. These values are estimates based on average measured power rather than continuous measurements of energy over an entire day, and they depend on the assumption that the measured averages remain representative throughout the selected period.

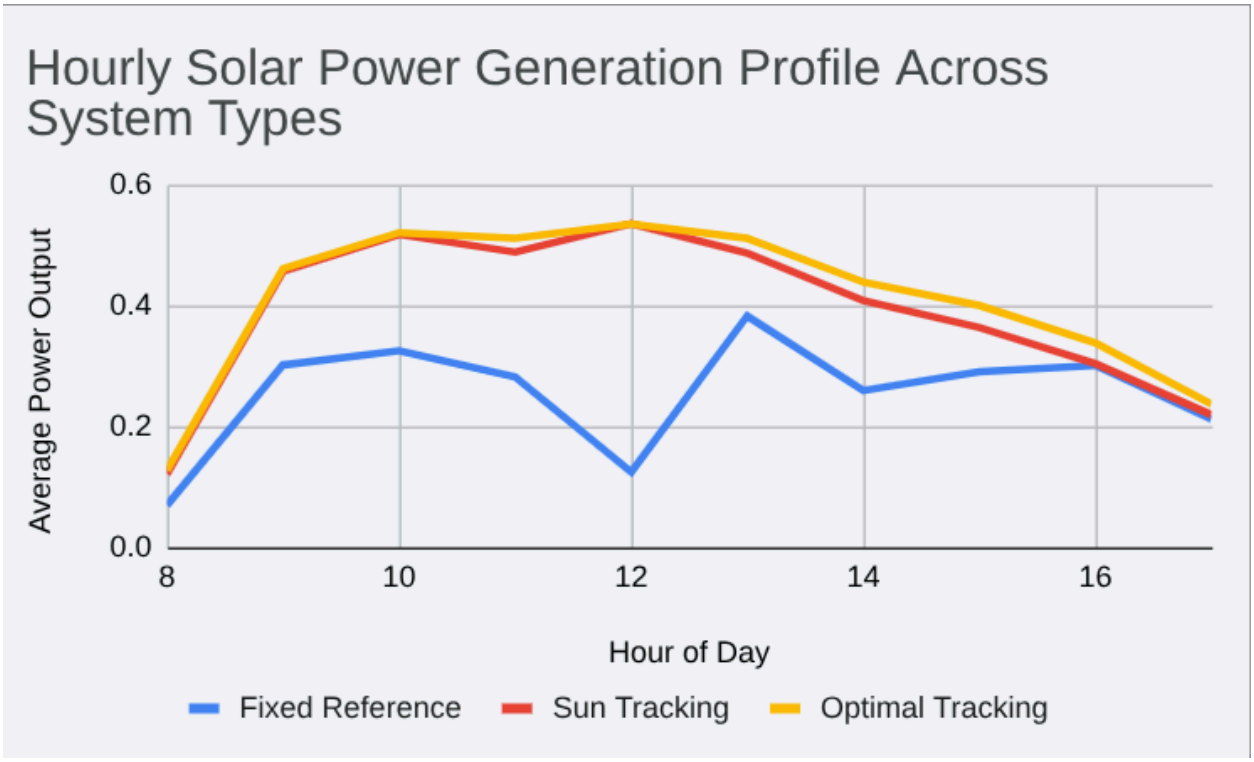


This graph compares power output from the fixed reference direction (blue), Sun direction (red), and optimal direction (orange).

The Sun direction and optimal direction curves generally produce higher power output than the fixed reference curve. The optimal-direction curve is slightly above the Sun direction curve at several points across the curve, suggesting that small adjustments around the calculated solar position may improve measured output at larger scales. However, the improvement is not consistent at every measurement point, and there are many sharp changes in the graph reflecting changing sunlight, variability in measurement, or positioning issues.



This graph compares average power output for three fixed reference observation points, three optimized tracking observation points, and three Sun tracking observation points.



Energy Input

The energy required to move the panel must also be considered when evaluating the usefulness of the tracking system. Each servo movement was estimated to take approximately one second, and a nominal power rating of 1 W per motor was used to illustrate the possible energy requirement. Under this assumption, energy consumed during motor operation can be estimated by multiplying the motors' power consumption by their total operating time. However, the nominal rating does not establish the actual power drawn by the servos, which may vary depending on whether they are moving, holding position, or operating under mechanical load. The current prototype did not independently measure servo energy consumption, so these estimates cannot establish the system's net energy gain. A more complete evaluation would measure the servos' actual voltage and current throughout the tracking sequence and subtract their energy consumption, along with that of other tracking electronics, from the additional energy produced by the panel.

Cost Analysis

The research presented shows that there is almost 80% increase in average power collected by photovoltaic panels repositioned directly towards the sun panel compared to rigid rooftop panels, and a 100% increase in average power collected by panels repositioned towards the optimal direction.

Scale

The data collected in this paper used a small 6 V photovoltaic panel, so the measured power output and the net power benefit should be interpreted within the scale of the prototype. The magnitude of this improvement may differ when applied to larger, commercial rooftop solar installations. Larger photovoltaic panels would produce substantially more electric power while taking in more energy to operate the trackking mechanism.

Bibliography:

https://www.energy.gov/cmei/systems/solar-photovoltaic-performance-and-efficiency-basics

Github code reference : https://github.com/emp0ry/solar-calculator/blob/main/main.py